

Table 66. I²C analog filter characteristics

Evaluated by characterization - Not tested in production.
 Measurement points are taken at 50% V_{DD}.

Symbol	Parameter	Min	Max	Unit
t _{AF}	Maximum pulse width of spikes that are suppressed by analog filter	50 ⁽¹⁾	160 ⁽²⁾	ns

1. Spikes with widths below t_{AF(min)} are filtered.
2. Spikes with widths above t_{AF(max)} are not filtered.

5.3.26

USART characteristics

Unless otherwise specified, the parameters given in Table 67 are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and V_{DD} supply voltage conditions summarized in not found, with the following configuration:

- Output speed set to OSPEEDRy[1:0] = 10
- Capacitive load C_L = 30 pF
- Measurement points done at 0.5 × V_{DD} level
- I/O compensation cell activated

Refer to I/O port characteristics for more details on the input/output alternate function characteristics (NSS, CK, TX, RX for USART).

Table 67. USART characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
f _{CK}	USART clock frequency	Master transmitter mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	18	
		Slave receiver mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	48	
		Slave transmitter mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	26	
t _{su(NSS)}	NSS setup time	Slave mode		t _{ker} ⁽¹⁾ + 2	-	-	ns
t _{h(NSS)}	NSS hold time	Slave mode		2	-	-	
t _{w(CKH)}	CK high and low time	Master mode		1/ f _{CK} /2-1	1/ f _{CK} /2	1/ f _{CK} /2+1	
t _{w(CKL)}							
t _{su(RX)}	Data input setup time	Master mode		18	-	-	
t _{su(RX)}		Slave mode		2.5	-	-	
t _{h(RX)}	Data input hold time	Master mode		0.5	-	-	
t _{h(RX)}		Slave mode		1	-	-	
t _{v(TX)}	Data output valid time	Slave mode	2.7 V ≤ V _{DD} ≤ 3.6 V	-	13.5	19	
		Master mode	2.7 V ≤ V _{DD} ≤ 3.6 V	-	2.5	3	
t _{h(TX)}	Data output hold time	Slave mode		10	-	-	
t _{h(TX)}	Data output hold time	Master mode		1	-	-	

1. T_{ker} is the usart_ker_ck_pres clock period.